a. Original IR image

b. With grayscale transform

FIGURE 23-13
Grayscale processing. Image (a) was acquired with an infrared camera in total darkness. Brightness in the image is related to the temperature, accounting for the appearance of the warm human body and the hot truck grill. Image (b) was processed with the manual grayscale transform shown in Fig. 23-14c.

The image in Fig. 23-13a was acquired in total darkness by using a CCD camera that is sensitive in the far infrared. The parameter being imaged is *temperature*: the hotter the object, the more infrared energy it emits and the brighter it appears in the image. This accounts for the background being very black (cold), the body being gray (warm), and the truck grill being white (hot). These systems are great for the military and police; you can see the other guy when he can't even see himself! The image in (a) is difficult to view because of the uneven distribution of pixel values. Most of the image is so dark that details cannot be seen in the scene. On the other end, the grill is near white saturation.

The histogram of this image is displayed in Fig. 23-14a, showing that the background, human, and grill have reasonably separate values. In this example, we will increase the contrast in the background and the grill, at the expense of everything else, including the human body. Figure (b) represents this strategy. We declare that the lowest pixel values, the background, will have a relative contrast of twelve. Likewise, the highest pixel values, the grill, will have a relative contrast of six. The body will have a relative contrast of one, with a staircase transition between the regions. All these values are determined by trial and error.

The grayscale transform resulting from this strategy is shown in (c), labeled *manual*. It is found by taking the running sum (i.e., the discrete integral) of the curve in (b), and then normalizing so that it has a value of 255 at the